

1.4 Circular measure

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

In radians, arc length $s=r\theta$ and sector area $A=\frac{1}{2}r^2\theta$. These formulas require θ in radians, not degrees. Convert θ° to $\theta\pi/180$; for a segment subtract triangle area $\frac{1}{2}r^2\sin\theta$ from sector area.

Worked example: follow the reasoning

For radius 6 cm and angle $\pi/3$, arc length is 2π cm and sector area is 6π cm².

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A sector has radius 5 cm and angle 1.2 rad. Find arc length and area.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$s=r\theta=6\text{ cm [1]}$; $A=\frac{1}{2}r^2\theta\text{ [1]}=15\text{ cm}^2\text{ [1]}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.